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EXPLORTARY OF DATA ANALYSIS

LAB ASSESSMENT -7

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REG NO : 23BDS0334

SLOT : L31+L32

COURSE CODE : BCSE331P

1. Load the dataset shared via email

AIM:

To load and inspect the given assessment.csv dataset.

CODE:



assessment_7.r*

Source on Save

```r
#23BDS0334
#ArjunMishra
# Install packages if required
install.packages("e1071")
install.packages("ggplot2")
install.packages("dplyr")
install.packages("reshape2")
install.packages("GGally")
# Load libraries
library(e1071)
library(ggplot2)
library(dplyr)
library(reshape2)
library(GGally)
# Load dataset
data <- read.csv("assessment.csv")
# View first few observations
head(data)
# View last few observations
tail(data)
# Check structure
str(data)
# Dimensions
dim(data)
# Column names
names(data)
# Summary
summary(data)
```

OUTPUT:

```
> # Load dataset
> data <- read.csv("assessment.csv")
> # View first few observations
> head(data)
  rownames Gunnel Time Fromlow Slope  Rw Amphiso Subst Pool Water Cobble
1        1      0  786     190    10 0.1       0     3    0     0      0
2        2      0  794     182    10 0.0       0     3    0     0      0
3        3      0  798     178    20 0.0       0     1    0     0      0
4        4      0  800     176    70 0.3       0     1    0     1      0
5        5      0  804     172    40 0.3       0     1    0     1      0
6        6      0  808     168    60 0.3       0     1    0     1      0
> # View last few observations
> tail(data)
     rownames Gunnel Time Fromlow Slope Rw Amphiso Subst Pool Water Cobble
1587     1587      0  778     130     0  1       0     1    0     1      0
1588     1588      0  782     134    10  1       0     1    0     0      0
1589     1589      0  784     136    10  1       0     1    1     1      0
1590     1590      0  790     142    10  1       0     1    0     1      0
1591     1591      0  793     145     0  1       1     7    0     1      1
1592     1592      0  796     148    20  1       0     1    1     1      0
> # Check structure
> str(data)
'data.frame':   1592 obs. of  11 variables:
 $ rownames: int  1 2 3 4 5 6 7 8 9 10 ...
 $ Gunnel  : int  0 0 0 0 0 0 0 0 0 0 ...
 $ Time    : int  786 794 798 800 804 808 814 817 821 826 ...
 $ Fromlow : int  190 182 178 176 172 168 162 159 155 150 ...
 $ Slope   : int  10 10 20 70 40 60 40 0 50 20 ...
 $ Rw      : num  0.1 0 0 0.3 0.3 0.3 0.3 0.9 0.5 0.1 ...
 $ Amphiso : int  0 0 0 0 0 0 0 1 0 0 ...
 $ Subst   : int  3 3 1 1 1 1 1 1 1 1 ...
 $ Pool    : int  0 0 0 0 0 0 0 1 0 0 ...
 $ Water   : int  0 0 0 1 1 1 1 1 1 0 ...
 $ Cobble  : int  0 0 0 0 0 0 0 0 0 0 ...
> # Dimensions
> dim(data)
[1] 1592   11
> # Column names
> names(data)
 [1] "rownames" "Gunnel"   "Time"     "Fromlow"  "Slope"    "Rw"       "Amphiso"  "Subst"    "Pool"     "Water"
[11] "Cobble"
> # Summary
> summary(data)
    rownames         Gunnel            Time         Fromlow          Slope             Rw             Amphiso
 Min.   :   1.0   Min.   :0.00000   Min.   :340.0   Min.   :  0.00   Min.   : 0.000   Min.   :0.0000   Min.   :0.0000
 1st Qu.: 398.8   1st Qu.:0.00000   1st Qu.:623.0   1st Qu.: 71.75   1st Qu.: 0.000   1st Qu.:0.0000   1st Qu.:0.0000
```

```
 $ Cobble  : int  0 0 0 0 0 0 0 0 0 0 ...
> # Dimensions
> dim(data)
[1] 1592   11
> # Column names
> names(data)
 [1] "rownames" "Gunnel"   "Time"     "Fromlow"  "Slope"    "Rw"       "Amphiso"  "Subst"    "Pool"     "Water"
[11] "Cobble"
> # Summary
> summary(data)
    rownames          Gunnel             Time          Fromlow          Slope             Rw             Amphiso
 Min.   :   1.0   Min.   :0.00000   Min.   :340.0   Min.   :  0.00   Min.   : 0.000   Min.   :0.0000   Min.   :0.0000
 1st Qu.: 398.8   1st Qu.:0.00000   1st Qu.:623.0   1st Qu.: 71.75   1st Qu.: 0.000   1st Qu.:0.0000   1st Qu.:0.0000
 Median : 796.5   Median :0.00000   Median :712.5   Median :158.00   Median : 0.000   Median :0.6000   Median :0.0000
 Mean   : 796.5   Mean   :0.03015   Mean   :705.2   Mean   :149.41   Mean   : 9.504   Mean   :0.5271   Mean   :0.4642
 3rd Qu.:1194.2   3rd Qu.:0.00000   3rd Qu.:810.0   3rd Qu.:224.25   3rd Qu.:10.000   3rd Qu.:1.0000   3rd Qu.:1.0000
 Max.   :1592.0   Max.   :1.00000   Max.   :983.0   Max.   :304.00   Max.   :90.000   Max.   :1.0000   Max.   :4.0000
     Subst             Pool             Water            Cobble
 Min.   : 1.000   Min.   :0.0000   Min.   :0.0000   Min.   :0.000
 1st Qu.: 1.000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.000
 Median : 2.000   Median :0.0000   Median :1.0000   Median :0.000
 Mean   : 3.106   Mean   :0.2023   Mean   :0.5088   Mean   :0.191
 3rd Qu.: 4.000   3rd Qu.:0.0000   3rd Qu.:1.0000   3rd Qu.:0.000
 Max.   :13.000   Max.   :1.0000   Max.   :1.0000   Max.   :1.000
>
```

RESULT:

The dataset was successfully loaded and its structure and basic characteristics were examined.

2. **Compute summary statistics: Mean, Median, Minimum, Maximum, Range, Variance, Standard Deviation, Interquartile Range (IQR), Skewness (using e1071::skewness), and Kurtosis (using e1071::kurtosis).**

AIM:

To Compute summary statistics: Mean, Median, Minimum, Maximum, Range, Variance, Standard Deviation, Interquartile Range (IQR), Skewness (using e1071::skewness), and Kurtosis (using e1071::kurtosis)

CODE:

assessment_7.r* ×

Source on Save

```
#23BDS0334
#ArjunMishra
# Select numerical variables
num_data <- data[, c("Time", "Fromlow", "Slope", "Rw")]
# Calculate statistics
summary_stats <- data.frame(
  Variable = names(num_data),
  Mean = sapply(num_data, mean, na.rm = TRUE),
  Median = sapply(num_data, median, na.rm = TRUE),
  Minimum = sapply(num_data, min, na.rm = TRUE),
  Maximum = sapply(num_data, max, na.rm = TRUE),
  Range = sapply(num_data, function(x) diff(range(x, na.rm = TRUE))),
  Variance = sapply(num_data, var, na.rm = TRUE),
  SD = sapply(num_data, sd, na.rm = TRUE),
  IQR = sapply(num_data, IQR, na.rm = TRUE),
  Skewness = sapply(num_data, skewness, na.rm = TRUE),
  Kurtosis = sapply(num_data, kurtosis, na.rm = TRUE)
)
print(summary_stats)
```

OUTPUT:

```
> #ArjunMishra
> # Select numerical variables
> num_data <- data[, c("Time", "Fromlow", "Slope", "Rw")]
> # Calculate statistics
> summary_stats <- data.frame(
+   Variable = names(num_data),
+   Mean = sapply(num_data, mean, na.rm = TRUE),
+   Median = sapply(num_data, median, na.rm = TRUE),
+   Minimum = sapply(num_data, min, na.rm = TRUE),
+   Maximum = sapply(num_data, max, na.rm = TRUE),
+   Range = sapply(num_data, function(x) diff(range(x, na.rm = TRUE))),
+   Variance = sapply(num_data, var, na.rm = TRUE),
+   SD = sapply(num_data, sd, na.rm = TRUE),
+   IQR = sapply(num_data, IQR, na.rm = TRUE),
+   Skewness = sapply(num_data, skewness, na.rm = TRUE),
+   Kurtosis = sapply(num_data, kurtosis, na.rm = TRUE)
+ )
> print(summary_stats)
         Variable        Mean Median Minimum Maximum Range     Variance          SD   IQR    Skewness   Kurtosis
Time         Time 705.2437186  712.5     340     983   643 1.845621e+04 135.8536375 187.0 -0.39280307 -0.2901323
Fromlow   Fromlow 149.4089196  158.0       0     304   304 6.833060e+03  82.6623264 152.5 -0.06568938 -1.2290229
Slope       Slope   9.5037688    0.0       0      90    90 2.524877e+02  15.8898625  10.0  2.95020038 10.2046273
Rw             Rw   0.5270729    0.6       0       1     1 1.774753e-01   0.4212781   1.0 -0.14354354 -1.7260733
>
```

RESULT:

A structured data frame containing all required descriptive statistics is produced for: (Time From low Slope Rw)

3. **Store these values in a single structured data.frame or named vector and print it.**

AIM:

To Store these values in a single structured data.frame or named vector and print it.

CODE:

assessment_7.r*

Source on Save

```
#23BDS0334
#ArjunMishra
print(summary_stats)
# Round values for easy presentation
summary_stats[,-1] <- round(summary_stats[,-1], 3)
print(summary_stats)
```

OUTPUT:

Console | Background Jobs

R 4.5.2 · ~/

```
> #23BDS0334
> #ArjunMishra
> print(summary_stats)
       Variable        Mean Median Minimum Maximum Range     Variance          SD   IQR    Skewness   Kurtosis
Time       Time 705.2437186  712.5     340     983   643 1.845621e+04 135.8536375 187.0 -0.39280307 -0.2901323
Fromlow Fromlow 149.4089196  158.0       0     304   304 6.833060e+03  82.6623264 152.5 -0.06568938 -1.2290229
Slope     Slope   9.5037688    0.0       0      90    90 2.524877e+02  15.8898625  10.0  2.95020038 10.2046273
Rw           Rw   0.5270729    0.6       0       1     1 1.774753e-01   0.4212781   1.0 -0.14354354 -1.7260733
> # Round values for easy presentation
> summary_stats[,-1] <- round(summary_stats[,-1], 3)
> print(summary_stats)
       Variable    Mean Median Minimum Maximum Range  Variance      SD   IQR Skewness Kurtosis
Time       Time 705.244  712.5     340     983   643 18456.211 135.854 187.0   -0.393   -0.290
Fromlow Fromlow 149.409  158.0       0     304   304  6833.060  82.662 152.5   -0.066   -1.229
Slope     Slope   9.504    0.0       0      90    90   252.488  15.890  10.0    2.950   10.205
Rw           Rw   0.527    0.6       0       1     1     0.177   0.421   1.0   -0.144   -1.726
>
```

RESULT:

The calculated descriptive statistics are stored in a single structured data.frame.

4. **Create a frequency distribution table and a relative frequency distribution table for a categorical variable. Generate a Pie Chart and a Bar Plot to represent the distribution.**

AIM:

To Create a frequency distribution table and a relative frequency distribution table for a categorical variable. Generate a Pie Chart and a Bar Plot to represent the distribution.

CODE:

assessment_7.r* ×

Source on Save

```r
#23BDS0334
#ArjunMishra
# Frequency distribution
freq_table <- table(data$Gunnel)
print(freq_table)
# Relative frequency
relative_freq <- prop.table(freq_table)
print(relative_freq)
# Percentage
percentage <- prop.table(freq_table) * 100
print(percentage)
# Pie Chart
pie(
  freq_table,
  main = "Distribution of Gunnel",
  labels = paste(names(freq_table),
                 round(percentage, 2), "%"))
# Bar Plot
barplot(
  freq_table,
  main = "Frequency Distribution of Gunnel",
  xlab = "Gunnel",
  ylab = "Frequency")
```

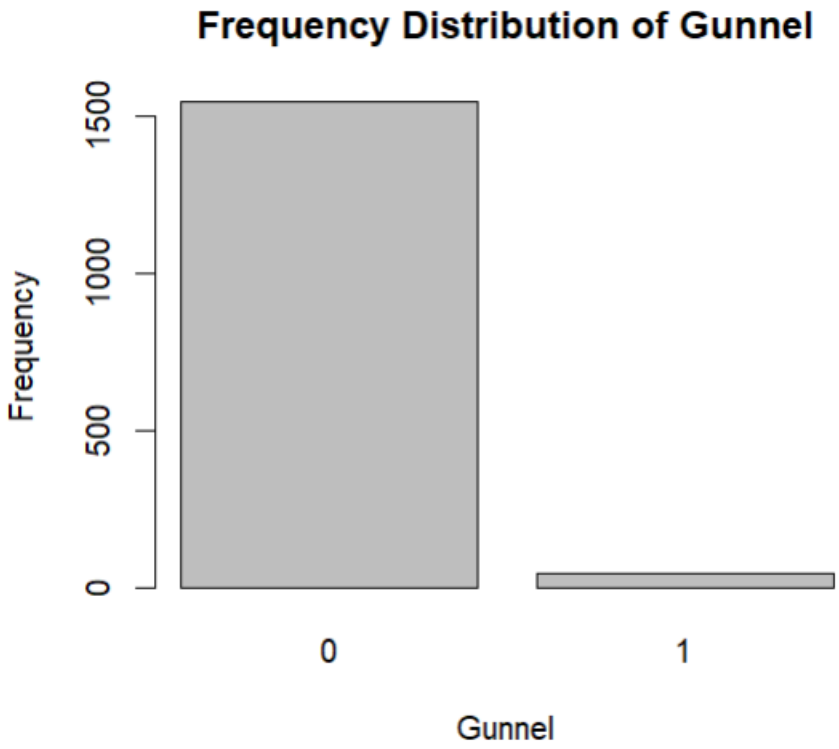
OUTPUT:

```
> #23BDS0334
> #ArjunMishra
> # Frequency distribution
> freq_table <- table(data$Gunnel)
> print(freq_table)

   0    1
1544   48
> # Relative frequency
> relative_freq <- prop.table(freq_table)
> print(relative_freq)

        0         1
0.96984925 0.03015075
> # Percentage
> percentage <- prop.table(freq_table) * 100
> print(percentage)

       0        1
96.984925  3.015075
> # Pie Chart
> pie(
+   freq_table,
+   main = "Distribution of Gunnel",
+   labels = paste(names(freq_table),
+                  round(percentage, 2), "%"))
> # Bar Plot
> barplot(
+   freq_table,
+   main = "Frequency Distribution of Gunnel",
+   xlab = "Gunnel",
+   ylab = "Frequency")
>
```



RESULT:

The frequency and relative-frequency distributions of Gunnel are represented using a **Pie Chart** and **Bar Plot**.

5. **Create a frequency distribution table with bin intervals for a numerical variable. Plot its Histogram and a Box Plot to check for outliers.**

AIM:

To Create a frequency distribution table with bin intervals for a numerical variable. Plot its Histogram and a Box Plot to check for outliers.

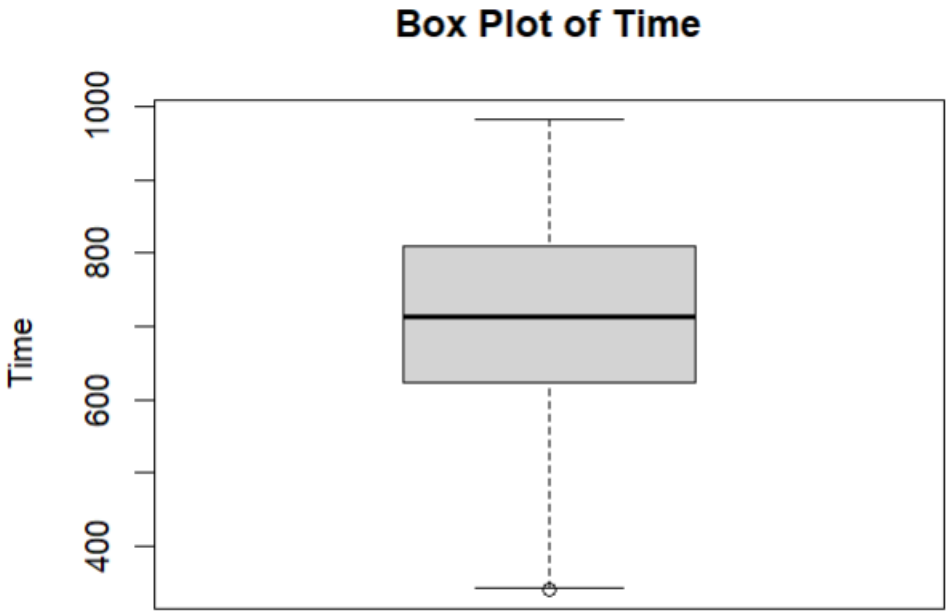
CODE:

assessment_7.r* | Source on Save

```r
#23BDS0334
#ArjunMishra
# Create bins
breaks <- seq(
  floor(min(data$Time, na.rm = TRUE)),
  ceiling(max(data$Time, na.rm = TRUE)),
  length.out = 11
)
# Frequency distribution with intervals
time_freq <- cut(
  data$Time,
  breaks = breaks,
  include.lowest = TRUE
)
freq_table_time <- table(time_freq)
print(freq_table_time)
# Histogram
hist(
  data$Time,
  breaks = 10,
  main = "Histogram of Time",
  xlab = "Time",
  ylab = "Frequency"
)
# Box Plot
boxplot(
  data$Time,
  main = "Box Plot of Time",
  ylab = "Time"
)
```

OUTPUT:

```
> #23BDS0334
> #ArjunMishra
> # Create bins
> breaks <- seq(
+   floor(min(data$Time, na.rm = TRUE)),
+   ceiling(max(data$Time, na.rm = TRUE)),
+   length.out = 11
+ )
> # Frequency distribution with intervals
> time_freq <- cut(
+   data$Time,
+   breaks = breaks,
+   include.lowest = TRUE
+ )
> freq_table_time <- table(time_freq)
> print(freq_table_time)
time_freq
[340,404] (404,469] (469,533] (533,597] (597,662] (662,726] (726,790] (790,854] (854,919] (919,983]
       33        85        61       119       318       212       271       314       102        77
> # Histogram
> hist(
+   data$Time,
+   breaks = 10,
+   main = "Histogram of Time",
+   xlab = "Time",
+   ylab = "Frequency"
+ )
> # Box Plot
> boxplot(
+   data$Time,
+   main = "Box Plot of Time",
+   ylab = "Time"
+ )
```



RESULT:

The histogram displays the distribution of Time, while the box plot helps identify possible **outliers**.

6. **Choose two categorical variables: Construct a Contingency Table (Cross-tabulation) and compute the row/column percentages using prop.table()**

AIM:

To Choose two categorical variables: Construct a Contingency Table (Cross-tabulation) and compute the row/column percentages using prop.table()

CODE:

assessment_7.r* ×

Source on Save

```
#23BDS0334
#ArjunMishra
# Contingency table
cross_table <- table(data$Gunnel, data$Water)
print(cross_table)
# Row percentages
row_percent <- prop.table(cross_table, margin = 1) * 100
print(row_percent)
# Column percentages
column_percent <- prop.table(cross_table, margin = 2) * 100
print(column_percent)
# Overall percentages
overall_percent <- prop.table(cross_table) * 100
print(overall_percent)
```

OUTPUT:

```
> #23BDS0334
> #ArjunMishra
> # Contingency table
> cross_table <- table(data$Gunnel, data$Water)
> print(cross_table)

     0   1
  0 770 774
  1  12  36
> # Row percentages
> row_percent <- prop.table(cross_table, margin = 1) * 100
> print(row_percent)

          0        1
  0 49.87047 50.12953
  1 25.00000 75.00000
> # Column percentages
> column_percent <- prop.table(cross_table, margin = 2) * 100
> print(column_percent)

           0         1
  0 98.465473 95.555556
  1  1.534527  4.444444
> # Overall percentages
> overall_percent <- prop.table(cross_table) * 100
> print(overall_percent)

            0          1
  0 48.3668342 48.6180905
  1  0.7537688  2.2613065
```

RESULT:

The contingency table shows the relationship between Gunnel and Water.

7. Create a Stacked Bar Plot or Grouped Bar Plot to visualize the relationship.

AIM:

To Create a Stacked Bar Plot or Grouped Bar Plot to visualize the relationship.

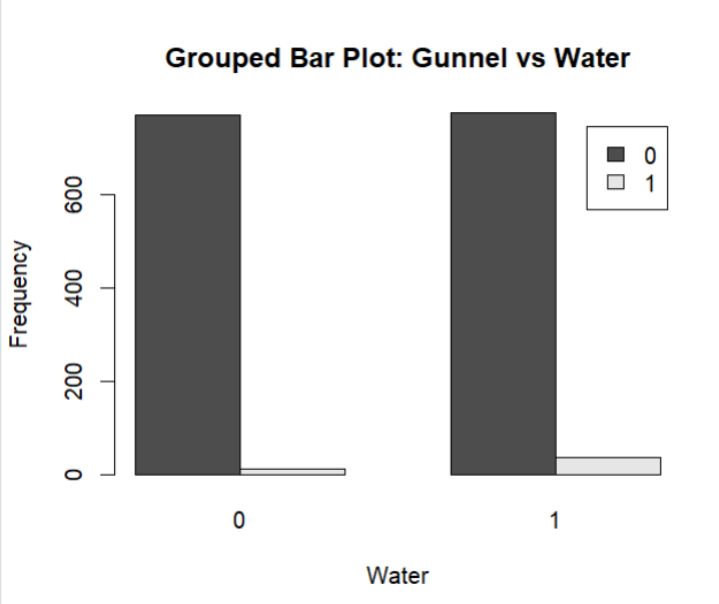
CODE:

assessment_7.r* ×

Source on Save

```r
#23BDS0334
#ArjunMishra
# Stacked bar plot
barplot(
  cross_table,
  beside = FALSE,
  main = "Gunnel vs Water",
  xlab = "Water",
  ylab = "Frequency",
  legend = rownames(cross_table)
)
# Grouped bar plot
barplot(
  cross_table,
  beside = TRUE,
  main = "Grouped Bar Plot: Gunnel vs Water",
  xlab = "Water",
  ylab = "Frequency",
  legend = rownames(cross_table)
)
```

OUTPUT:



RESULT:

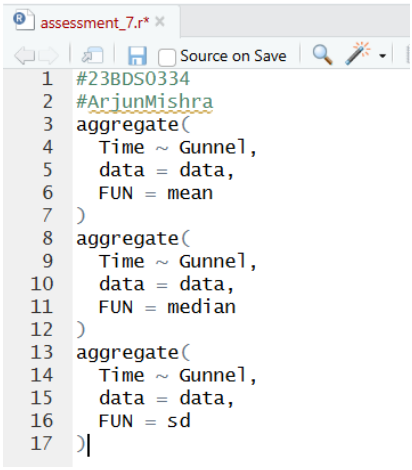
The bar plots visually represent the relationship between the two categorical variables.

8. **Select one categorical variable and one numerical variable: Compute the mean, median, and standard deviation of the numerical variable grouped by each category using aggregate()**

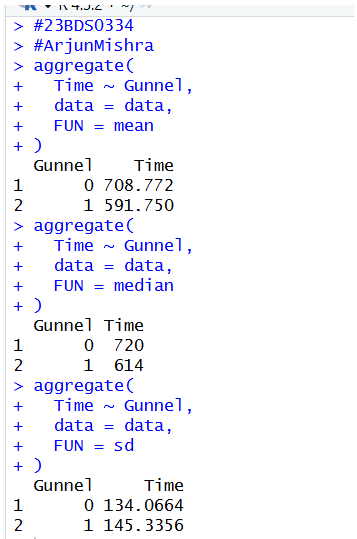
AIM:

To Select one categorical variable and one numerical variable: Compute the mean, median, and standard deviation of the numerical variable grouped by each category using aggregate()

CODE:



OUTPUT:



RESULT:

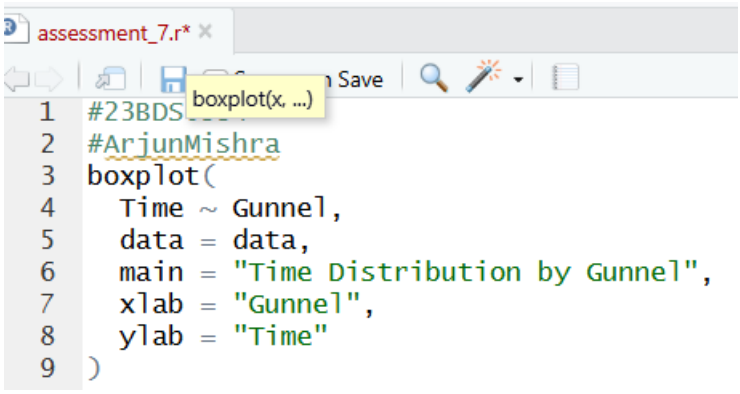
The mean, median and standard deviation of Time are calculated separately for each Gunnel category.

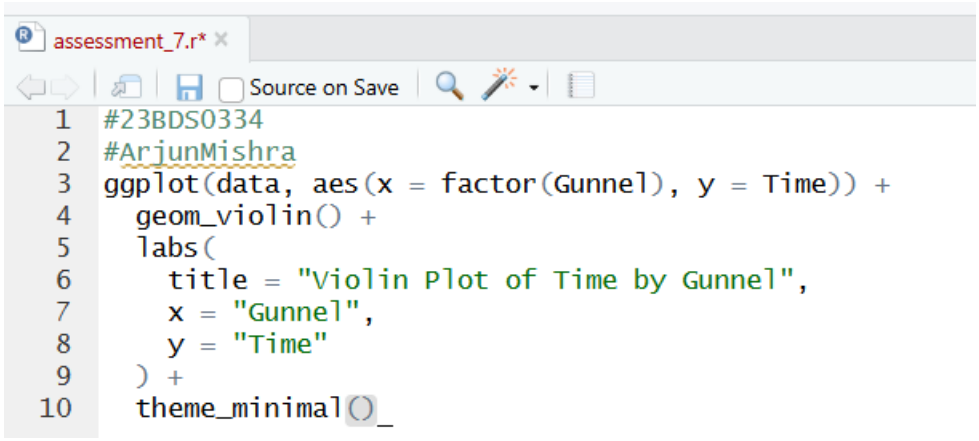
9. Generate Side-by-Side Box Plots and a Violin Plot to visualize the numerical distribution across categories.

AIM:

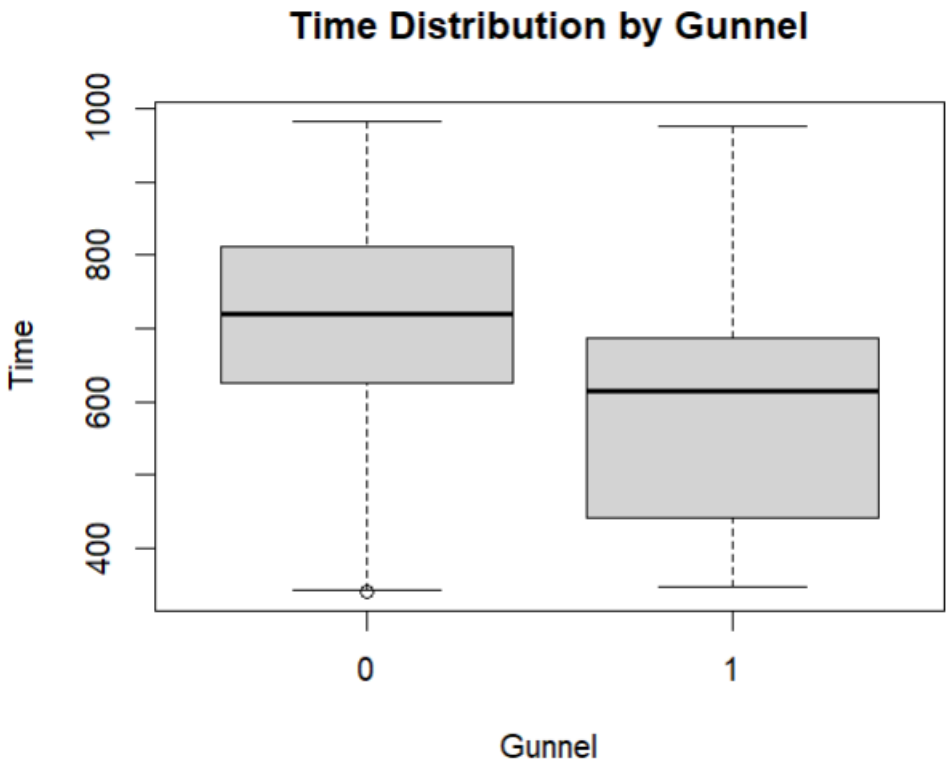
To Generate Side-by-Side Box Plots and a Violin Plot to visualize the numerical distribution across categories.

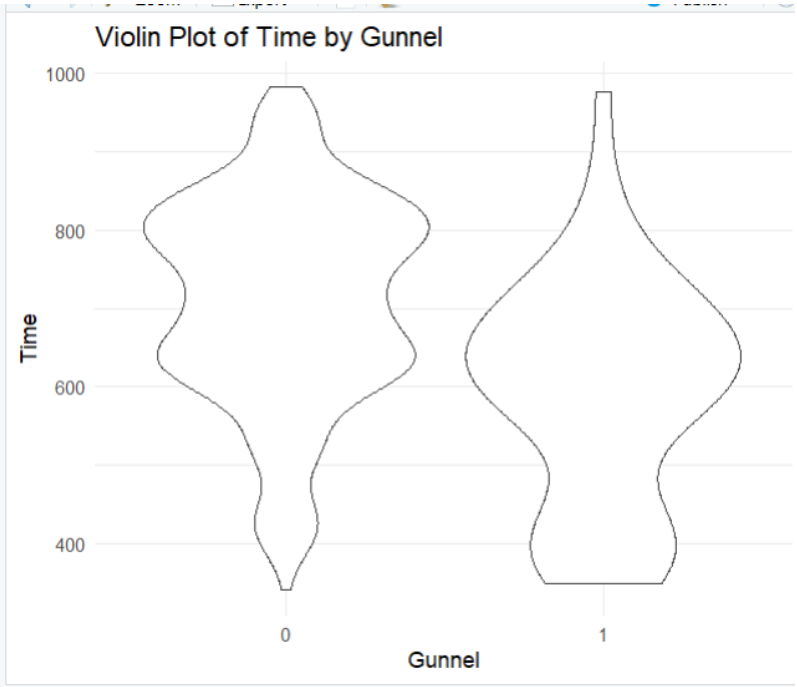
CODE:





OUTPUT:





RESULT:

The box plot and violin plot show the distribution, spread and possible outliers of Time for each Gunnel category.

10. Select two numerical variables: Calculate the Pearson and Spearman Correlation Coefficients (cor()) and Construct a Scatter Plot with a fitted linear regression trend line.

AIM:

To Select two numerical variables: Calculate the Pearson and Spearman Correlation Coefficients (cor()) and Construct a Scatter Plot with a fitted linear regression trend line.

CODE:

assessment_7.r* ×

Source on Save

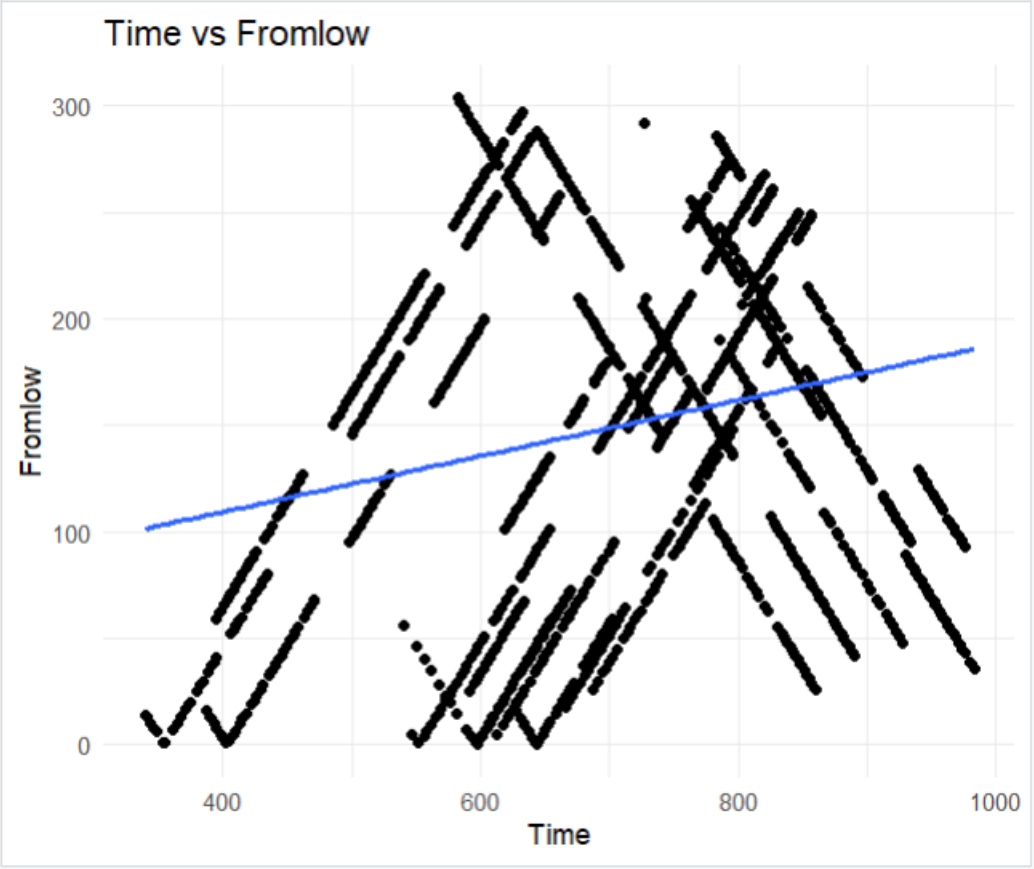
```r
#23BDS0334
#ArjunMishra
pearson_corr <- cor(
  data$Time,
  data$Fromlow,
  method = "pearson",
  use = "complete.obs"
)
print(pearson_corr)
spearman_corr <- cor(
  data$Time,
  data$Fromlow,
  method = "spearman",
  use = "complete.obs"
)

print(spearman_corr)
ggplot(data, aes(x = Time, y = Fromlow)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Time vs Fromlow",
    x = "Time",
    y = "Fromlow"
  ) +
  theme_minimal()
```

OUTPUT:

R 4.5.2 · ~/

```
> #23BDS0334
> #ArjunMishra
> pearson_corr <- cor(
+   data$Time,
+   data$Fromlow,
+   method = "pearson",
+   use = "complete.obs"
+ )
> print(pearson_corr)
[1] 0.2155932
> spearman_corr <- cor(
+   data$Time,
+   data$Fromlow,
+   method = "spearman",
+   use = "complete.obs"
+ )
>
> print(spearman_corr)
[1] 0.1811016
> ggplot(data, aes(x = Time, y = Fromlow)) +
+   geom_point() +
+   geom_smooth(method = "lm", se = FALSE) +
+   labs(
+     title = "Time vs Fromlow",
+     x = "Time",
+     y = "Fromlow"
+   ) +
+   theme_minimal()
`geom_smooth()` using formula = 'y ~ x'
>
```



RESULT:

Pearson and Spearman correlation coefficients measure the strength and direction of association between Time and Fromlow.

The scatter plot with a fitted linear regression line shows the relationship graphically.

11. **Select two categorical features and one numerical feature: Calculate summary statistics (Mean and Standard Deviation) for the numerical variable across all combinations of the two categorical variables and Create a Faceted Box Plot and a Grouped Bar Chart with Error Bars.**

CODE:

assessment_7.r* ×

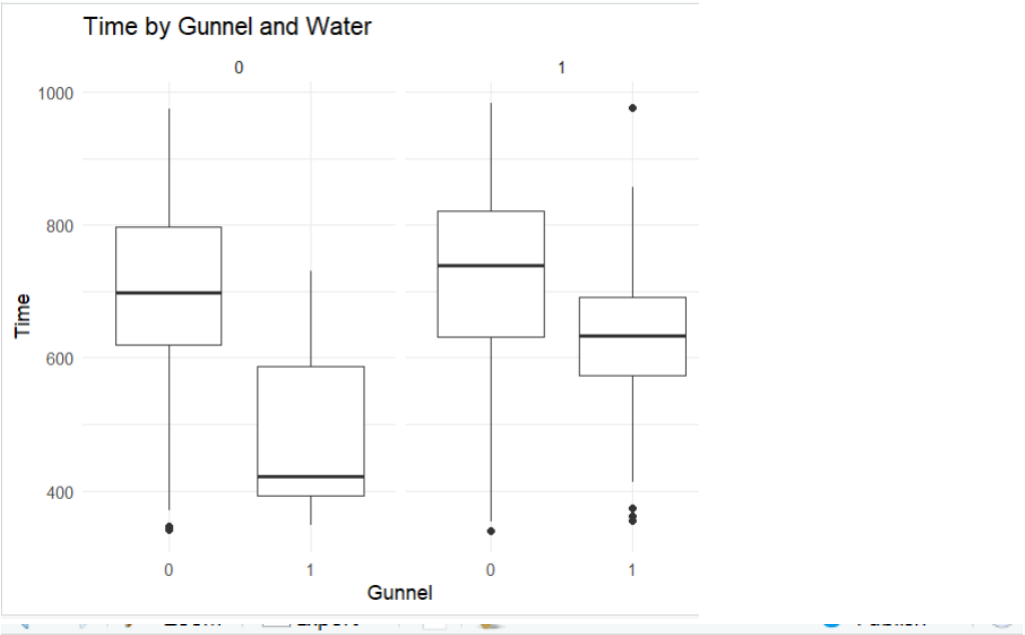
Source on Save

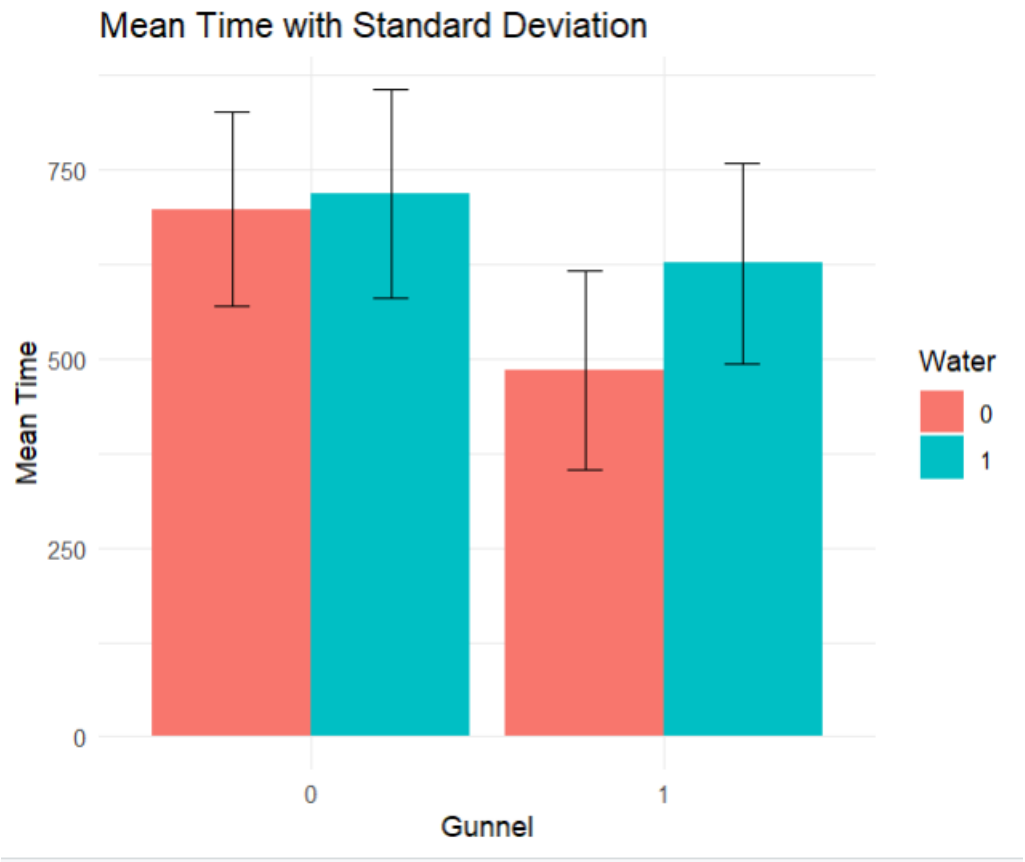
```r
#23BDS0334
#ArjunMishra
combination_stats <- data %>%
  group_by(Gunnel, Water) %>%
  summarise(
    Mean = mean(Time, na.rm = TRUE),
    SD = sd(Time, na.rm = TRUE),
    .groups = "drop"
  )
print(combination_stats)
ggplot(data, aes(x = factor(Gunnel), y = Time)) +
  geom_boxplot() +
  facet_wrap(~ Water) +
  labs(
    title = "Time by Gunnel and Water",
    x = "Gunnel",
    y = "Time"
  ) +
  theme_minimal()
```

OUTPUT:

R 4.5.2 · ~/

```
> #23BDS0334
> #ArjunMishra
> combination_stats <- data %>%
+   group_by(Gunnel, Water) %>%
+   summarise(
+     Mean = mean(Time, na.rm = TRUE),
+     SD = sd(Time, na.rm = TRUE),
+     .groups = "drop"
+   )
> print(combination_stats)
# A tibble: 4 × 4
  Gunnel Water  Mean    SD
   <int> <int> <dbl> <dbl>
1      0     0  699.  129.
2      0     1  719.  138.
3      1     0  485.  132.
4      1     1  627.  133.
> ggplot(data, aes(x = factor(Gunnel), y = Time)) +
+   geom_boxplot() +
+   facet_wrap(~ Water) +
+   labs(
+     title = "Time by Gunnel and Water",
+     x = "Gunnel",
+     y = "Time"
+   ) +
+   theme_minimal()
>
```





RESULT:

The analysis compares the mean and variability of Time across every combination of Gunnel and Water.

12. **Select one categorical variable and two numerical variables: Construct a Scatter Plot of the two numerical features where points are colored/shaped by the categorical feature. compute a correlation matrix between the two numerical variables separately for each category level.**

CODE:

Scatter Plot

assessment_7.r*

Source on Save

```r
#23BDS0334
#ArjunMishra
ggplot(
  data,
  aes(
    x = Time,
    y = Fromlow,
    color = factor(Gunnel),
    shape = factor(Gunnel)
  )
) +
  geom_point() +
  labs(
    title = "Time vs Fromlow by Gunnel",
    x = "Time",
    y = "Fromlow",
    color = "Gunnel",
    shape = "Gunnel"
  ) +
  theme_minimal()
```

Correlation Separately for Each Category

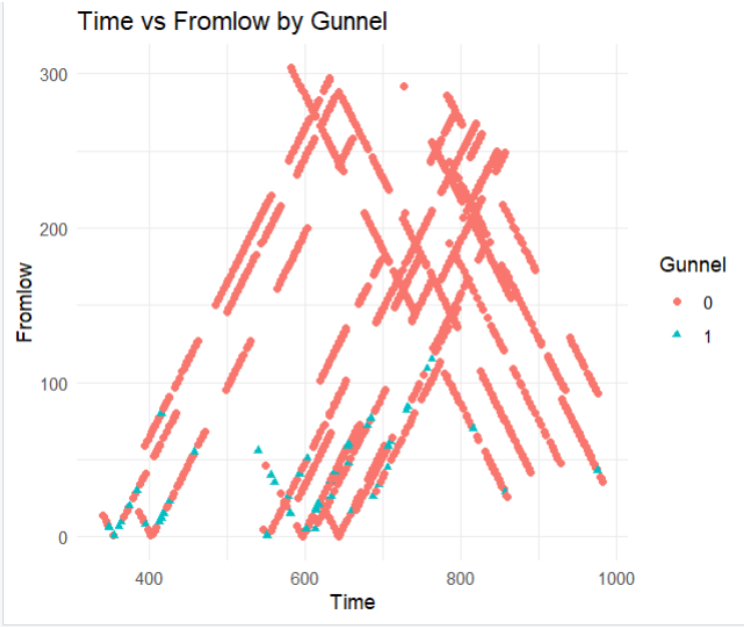
assessment_7.r*

Source on Save

```r
#23BDS0334
#ArjunMishra
cor_by_gunnel <- data %>%
  group_by(Gunnel) %>%
  summarise(
    Pearson = cor(
      Time,
      Fromlow,
      method = "pearson",
      use = "complete.obs"
    ),
    Spearman = cor(
      Time,
      Fromlow,
      method = "spearman",
      use = "complete.obs"
    )
  )

print(cor_by_gunnel)
```

OUTPUT:




```
> #23BDS0334
> #ArjunMishra
> cor_by_gunnel <- data %>%
+   group_by(Gunnel) %>%
+   summarise(
+     Pearson = cor(
+       Time,
+       Fromlow,
+       method = "pearson",
+       use = "complete.obs"
+     ),
+     Spearman = cor(
+       Time,
+       Fromlow,
+       method = "spearman",
+       use = "complete.obs"
+     )
+   )
>
> print(cor_by_gunnel)
# A tibble: 2 × 3
  Gunnel Pearson Spearman
   <int>   <dbl>    <dbl>
1      0   0.186    0.146
2      1   0.536    0.647
>
```

RESULT:

The scatter plot shows the relationship between Time and Fromlow separately for each Gunnel category.

Category-wise Pearson and Spearman correlations are also calculated.

13. Select at least 3 numerical features from the dataset: Generate a full Correlation Matrix and plot it as a Heatmap

AIM:

To Select at least 3 numerical features from the dataset: Generate a full Correlation Matrix and plot it as a Heatmap.

CODE:

assessment_7.r*

Source on Save

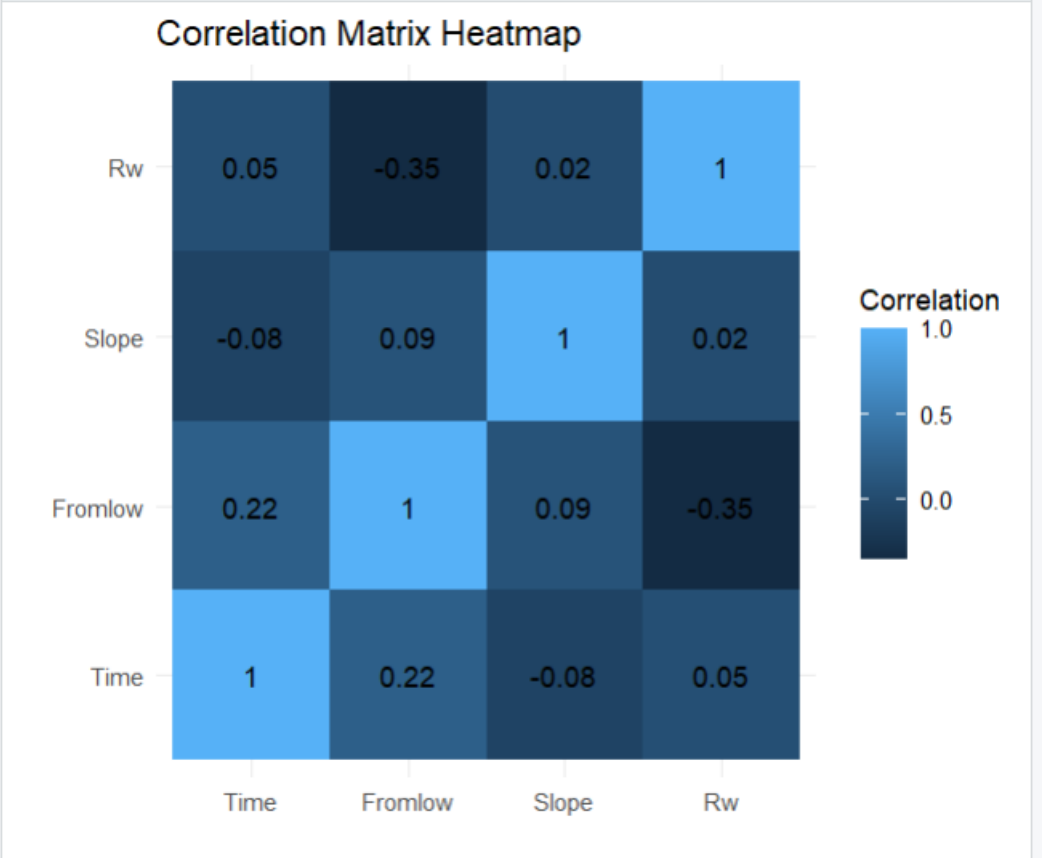
```
#23BDS0334
#ArjunMishra
cor_matrix <- cor(
  data[, c("Time", "Fromlow", "Slope", "Rw")],
  use = "complete.obs",
  method = "pearson"
)
print(cor_matrix)
cor_data <- melt(cor_matrix)

ggplot(
  cor_data,
  aes(x = Var1, y = Var2, fill = value)
) +
  geom_tile() +
  geom_text(aes(label = round(value, 2))) +
  labs(
    title = "Correlation Matrix Heatmap",
    x = "",
    y = "",
    fill = "Correlation"
  ) +
  theme_minimal()
```

OUTPUT:

R 4.5.2 · ~/

```
> #23BDS0334
> #ArjunMishra
> cor_matrix <- cor(
+   data[, c("Time", "Fromlow", "Slope", "Rw")],
+   use = "complete.obs",
+   method = "pearson"
+ )
> print(cor_matrix)
               Time    Fromlow       Slope          Rw
Time     1.00000000  0.2155932 -0.07692201  0.04781209
Fromlow  0.21559316  1.0000000  0.09058570 -0.34861156
Slope   -0.07692201  0.0905857  1.00000000  0.01956647
Rw       0.04781209 -0.3486116  0.01956647  1.00000000
> cor_data <- melt(cor_matrix)
>
> ggplot(
+   cor_data,
+   aes(x = Var1, y = Var2, fill = value)
+ ) +
+   geom_tile() +
+   geom_text(aes(label = round(value, 2))) +
+   labs(
+     title = "Correlation Matrix Heatmap",
+     x = "",
+     y = "",
+     fill = "Correlation"
+   ) +
+   theme_minimal()
>
```



RESULT:

The correlation matrix and heatmap show the strength and direction of relationships among the numerical variables.

14. Produce a Pairs Plot / Scatterplot Matrix to examine pairwise relationships and distributions simultaneously.

AIM:

To Produce a Pairs Plot / Scatterplot Matrix to examine pairwise relationships and distributions simultaneously.

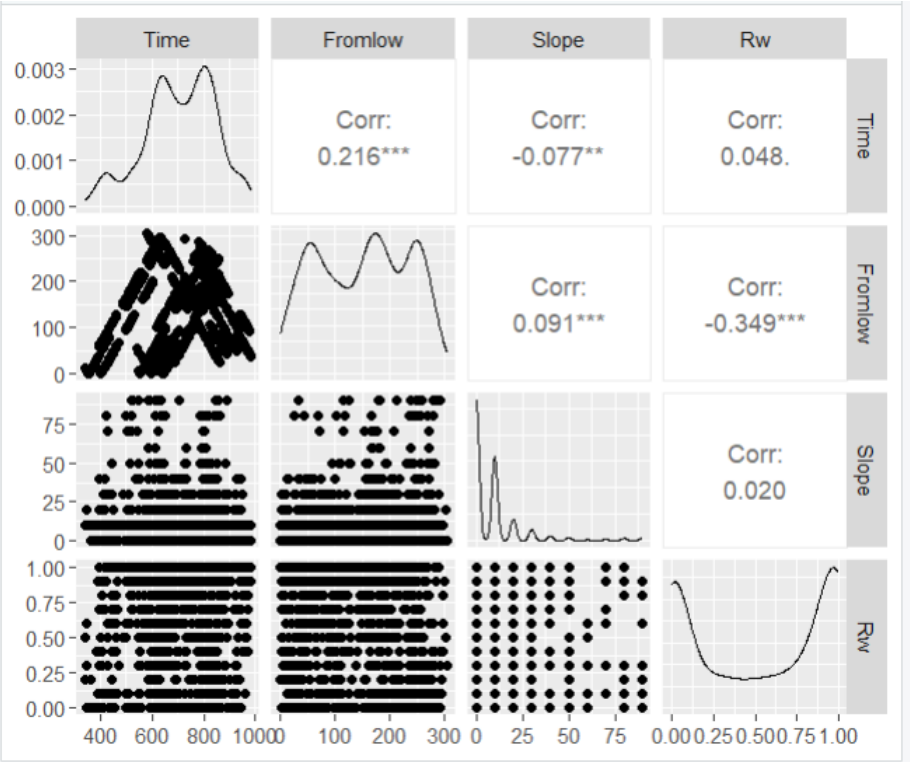
CODE:

assessment_7.r* ×

Source on Save

```
#23BDS0334
#ArjunMishra_
pairs(
  data[, c("Time", "Fromlow", "Slope", "Rw")],
  main = "Scatterplot Matrix"
)
ggpairs(
  data[, c("Time", "Fromlow", "Slope", "Rw")]
)
```

OUTPUT:



RESULT:

The pairs plot simultaneously displays:

- Pairwise scatter plots
- Distribution of individual variables
- Relationships between numerical variables

It is useful for identifying **correlation, trends, clusters and possible outliers**.